

lab-5

Yue Chenghao

227154

Question1

- code

```
java
import java.util.Scanner;

public class Question1 {
    public static void main(String[] args) {
        // define the month names and number of days in each month
        String[] months = {
            "January", "February", "March", "April", "May", "June",
            "July", "August", "September", "October", "November", "December"
        };
        int[] dom = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

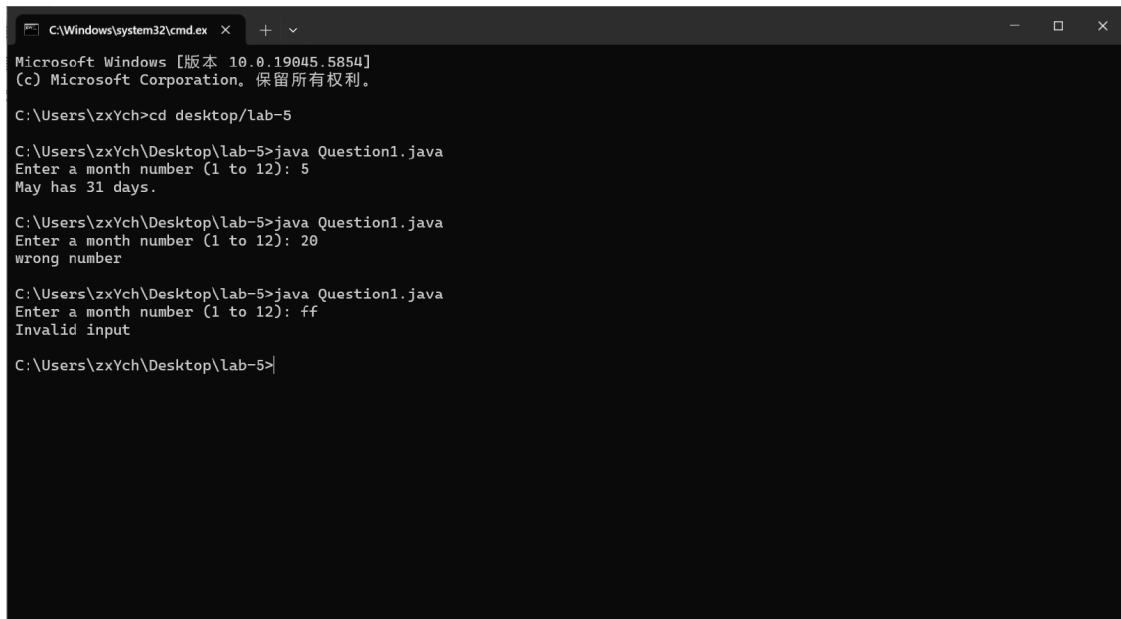
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a month number (1 to 12): ");

        try {
            int monthIndex = input.nextInt(); // read the month number from
            user input
            // indexing starts from 0, so we need to subtract 1 from the
            month number
            System.out.println(months[monthIndex - 1] + " has " +
            dom[monthIndex - 1] + " days.");
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("wrong number");
        } catch (Exception e) {
            System.out.println("Invalid input");
        }
        input.close();
    }
}
```

```
}
```

output



```
C:\Windows\system32\cmd.exe
Microsoft Windows [版本 10.0.19045.5854]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\zxYch>cd desktop/lab-5

C:\Users\zxYch\Desktop\lab-5>java Question1.java
Enter a month number (1 to 12): 5
May has 31 days.

C:\Users\zxYch\Desktop\lab-5>java Question1.java
Enter a month number (1 to 12): 20
wrong number

C:\Users\zxYch\Desktop\lab-5>java Question1.java
Enter a month number (1 to 12): ff
Invalid input

C:\Users\zxYch\Desktop\lab-5>
```

Question2

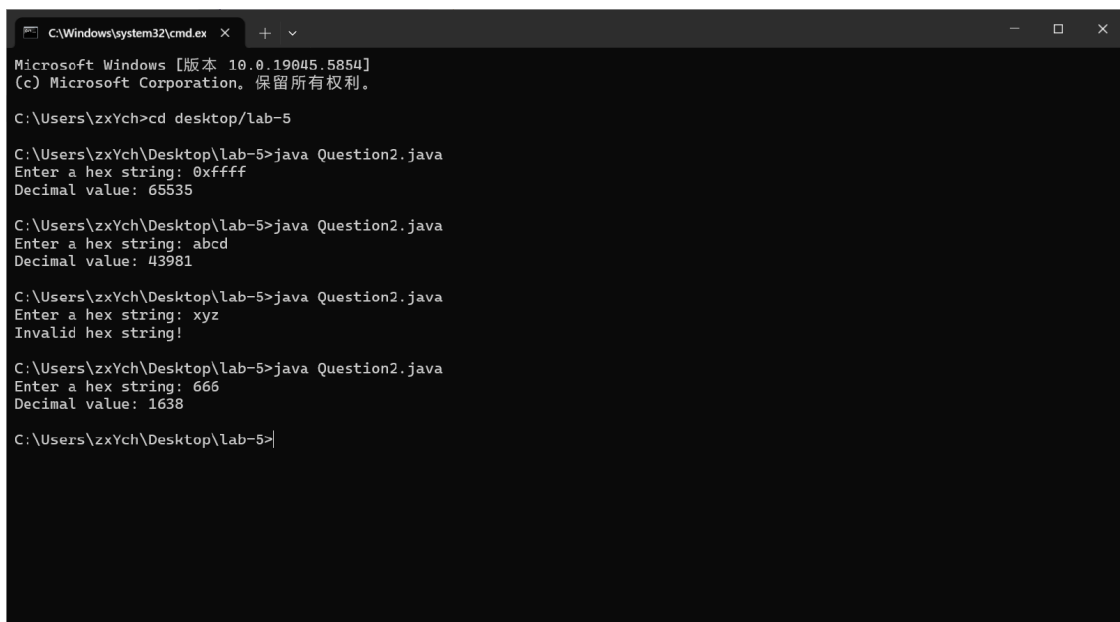
- code

```
import java.util.Scanner;

public class Question2 {
    // Convert hex string to decimal
    public static int hex2Dec(String hex) {
        // Remove any leading "0x" or "0X" if present
        if (hex.startsWith("0x") || hex.startsWith("0X")) {
            hex = hex.substring(2);
        }
        return Integer.parseInt(hex, 16);
    }
}
```

```
public static void main(String[] args) {  
    Scanner input = new Scanner(System.in);  
    System.out.print("Enter a hex string: ");  
    String userInput = input.nextLine();  
  
    try {  
        int decimal = hex2Dec(userInput);  
        System.out.println("Decimal value: " + decimal);  
    } catch (NumberFormatException e) {  
        System.out.println("Invalid hex string!");  
    }  
}
```

output



```
C:\Windows\system32\cmd.exe x + v  
Microsoft Windows [版本 10.0.19045.5854]  
(c) Microsoft Corporation. 保留所有权利。  
  
C:\Users\zxYch>cd desktop/lab-5  
  
C:\Users\zxYch\Desktop\lab-5>java Question2.java  
Enter a hex string: 0xffff  
Decimal value: 65535  
  
C:\Users\zxYch\Desktop\lab-5>java Question2.java  
Enter a hex string: abcd  
Decimal value: 43981  
  
C:\Users\zxYch\Desktop\lab-5>java Question2.java  
Enter a hex string: xyz  
Invalid hex string!  
  
C:\Users\zxYch\Desktop\lab-5>java Question2.java  
Enter a hex string: 666  
Decimal value: 1638  
  
C:\Users\zxYch\Desktop\lab-5>
```